

Employee Attrition Analysis

Identifying Factors Influencing Employee Turnover

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Dataset: [IBM HR Analytics Employee Attrition & Performance](#)
(1,470 employees x 35 features)

Tools: Python (pandas, numpy, matplotlib, seaborn, scipy) and SQL / BigQuery

Key business question: Which factors are most strongly associated with employee attrition, and what actions can HR take to improve retention?

1. Business problem

The company is losing too many employees. HR wants to understand who leaves most often, why people leave, which groups carry elevated risk, and which actions would improve retention. I approached this as a set of focused questions, answering each one with an aggregation, a chart, and (where it matters) a statistical test.

2. The data

Property	Value
Source	IBM HR Analytics Employee Attrition & Performance (Kaggle); fictional dataset by IBM
Format	One CSV file (WA_Fn-UseC_-HR-Employee-Attrition.csv)
Size	1,470 rows x 35 columns, ~1 MB in memory
Grain	One row per employee (a single snapshot in time)
Target	Attrition (Yes = left, No = stayed)
Quality	No missing values, no duplicates, unique IDs, no broken tenure values

Cleaning: dropped three constant columns (EmployeeCount, Over18, StandardHours) and the EmployeeNumber id. Note: PerformanceRating only ever takes the values 3 and 4, so it has almost no variation and was not used.

3. Method

- Data quality: checked for missing values, duplicates, constant columns and logical errors (YearsAtCompany never exceeds TotalWorkingYears).
- New columns: AttritionFlag (1/0, so the rate is a mean), TenureRatio (career share at the company), and buckets for age, tenure and income quartiles.
- SQL: the first exploration is done in SQL (SQLite in the notebook, with a full BigQuery version in sql/bigquery_queries.sql).
- Statistics: chi-square + Cramér's V for categories; Welch's t-test, Mann-Whitney and point-biserial correlation for numeric features; a two-proportion z-test (A/B test) for the gender question. Significance threshold $p < 0.05$.

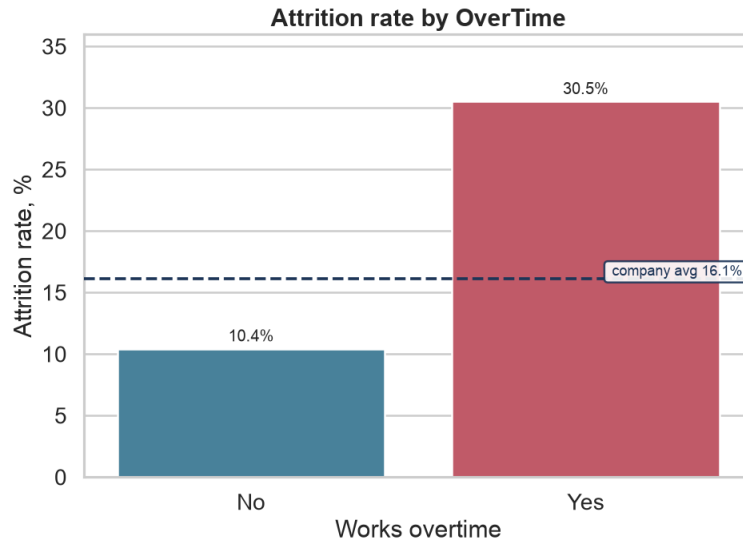
4. Analytical questions and results

No	Question	Result
1	Does overtime affect attrition?	Confirmed - strongest single factor
2	Is income related to turnover?	Confirmed - leavers earn ~38% less (median)
3	Which departments / roles lose the most?	Confirmed - Sales and Sales Reps
4	Do newer employees leave more?	Confirmed - the first year is riskiest
5	Do satisfaction and work-life balance matter?	Confirmed - and they stack
6	Does age matter?	Confirmed - younger people leave more
7	Marital status / business travel?	Confirmed
8	Do stock options and job level protect retention?	Confirmed
9	Real difference between men and women? (A/B test)	Rejected ($p = 0.26$)
10	Do risk factors compound?	Confirmed - worst combo ~8x the safest

5. Detailed findings

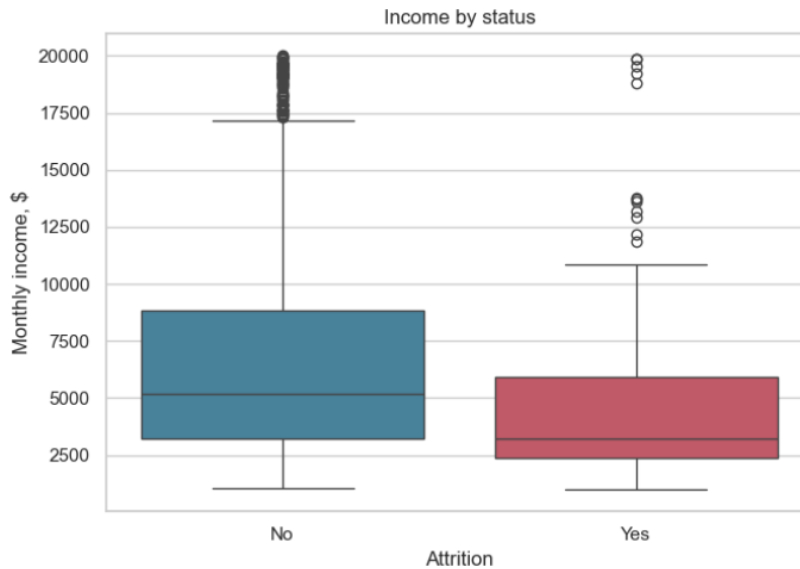
5.1 Overtime is the number-one driver

People who work overtime leave 30.5% of the time versus 10.4% of everyone else - almost three times as likely. The chi-square test confirms the gap is not chance (p around $1e-20$), and Cramér's V (0.24) is the strongest of any single factor.



5.2 Pay matters, but doesn't explain everything

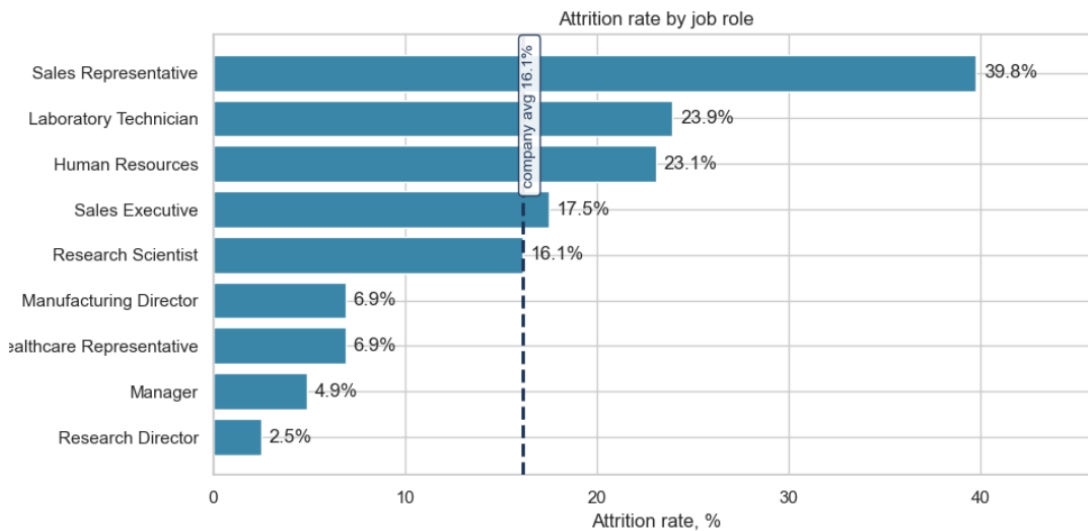
Leavers earn a median of about \$3,200 versus \$5,200 for those who stay (about 38% less). The difference is highly significant, but the correlation with attrition is only moderate (-0.16), so low pay raises risk mostly in combination with other factors (see 6.7).





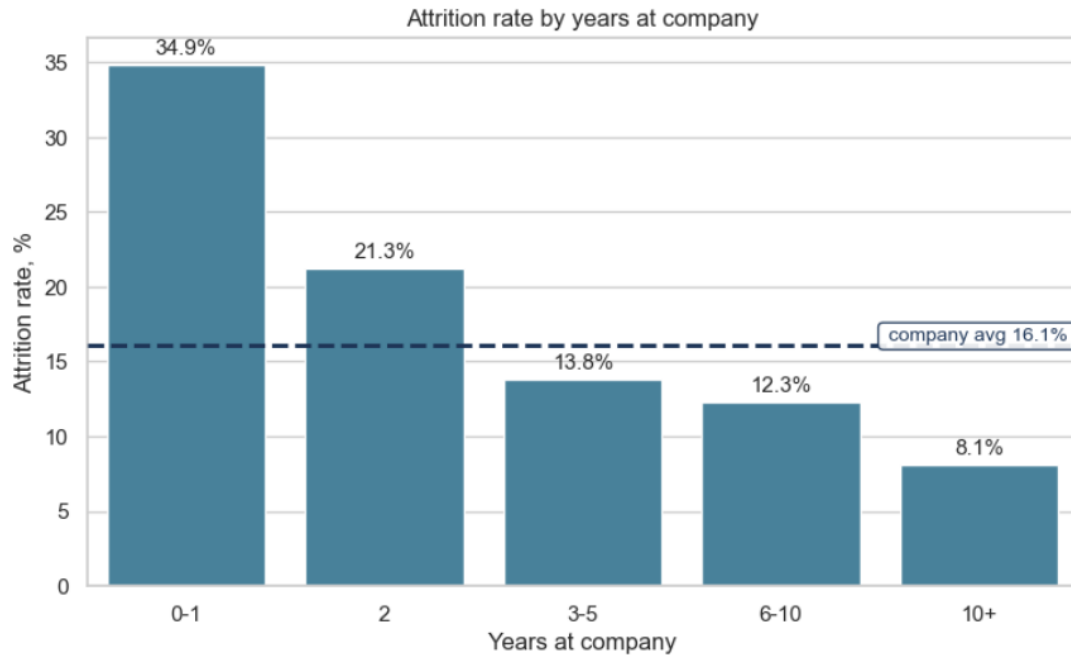
5.3 Risk concentrates in front-line and junior roles

By department Sales (20.6%) and HR (19.0%) are above average. By role the gap is sharper: Sales Representatives leave 39.8% of the time, versus 2.5% for Research Directors. It is a role problem, not a whole-department one.



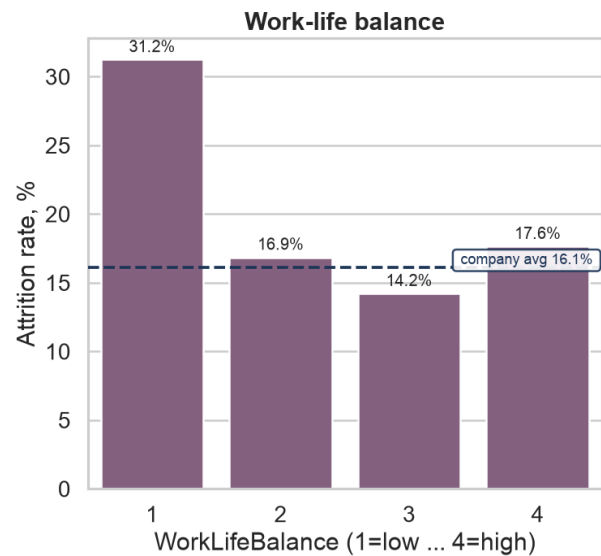
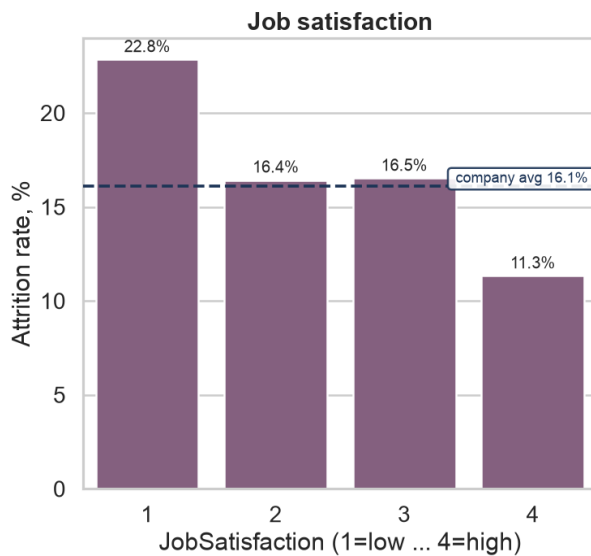
5.4 The first year is critical

34.9% of people leave in their first year, dropping to 8.1% among those with 10+ years. The onboarding window decides a lot - this is where retention effort pays off most.



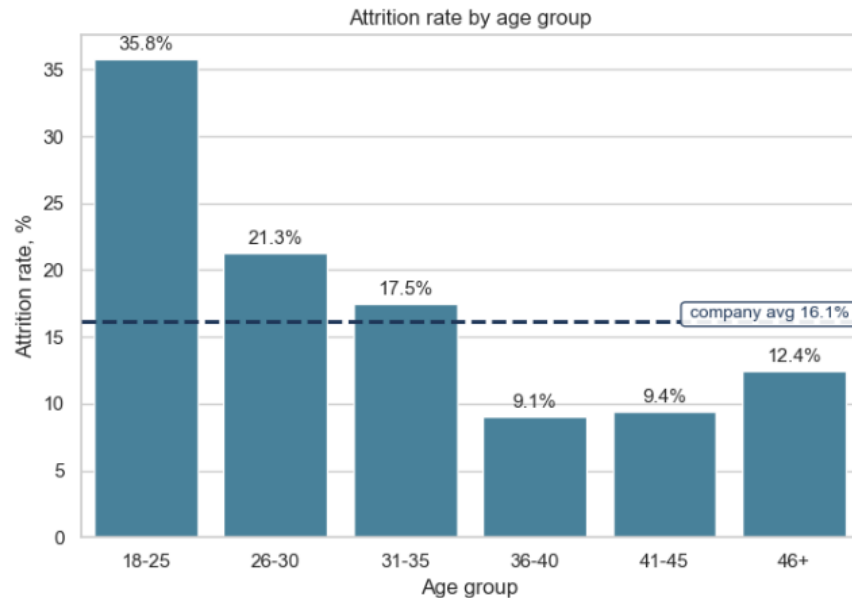
5.5 Satisfaction and work-life balance stack

Low job satisfaction (22.8% at level 1) and poor work-life balance (about 31% at level 1) both raise risk, and they combine: the worst cell of the pivot (satisfaction 1 and work-life balance 1) reaches 47.1%.



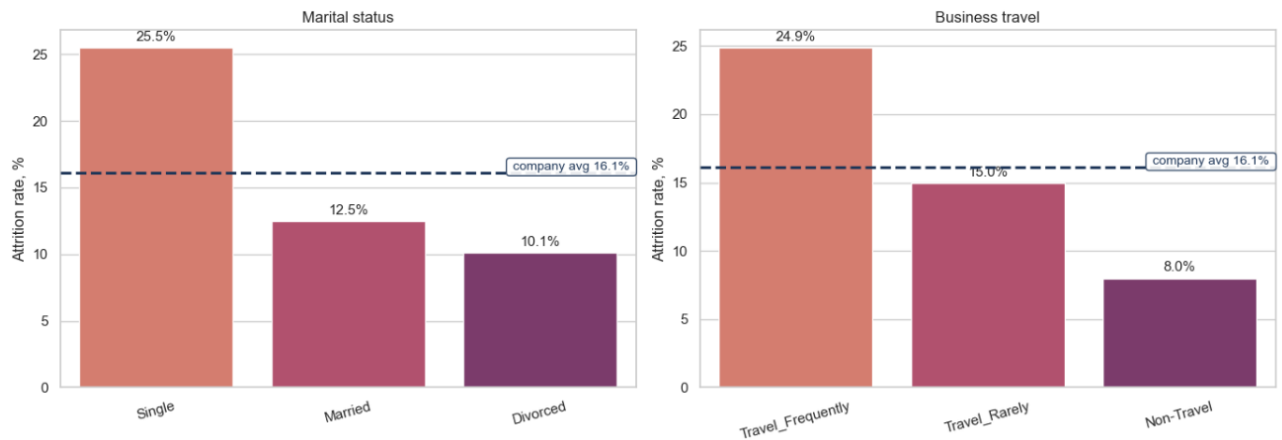
5.6 Younger employees are at the highest risk

Attrition is highest among the youngest employees: the 18–25 group leaves at 35.8%, which is more than double the company average of 16.1%. The rate then drops with age: 21.3% for ages 26–30, 17.5% for 31–35, and around 9–12% after age 36. This suggests that age matters, but mostly as a proxy for career stage, tenure, and seniority: younger and likely more junior employees are at the highest risk of leaving.



5.7. Single employees and frequent travelers need extra attention

Single employees show the highest attrition rate at **25.5%**, which is well above the company average of **16.1%** and roughly double the rate for married employees (**12.5%**). Divorced employees have the lowest rate at **10.1%**. Business travel shows a similar risk pattern: employees who travel frequently leave at **24.9%**, compared with **15.0%** for those who travel rarely and only **8.0%** for non-travellers. Overall, being single and frequent business travel both point to higher attrition risk, although the “single” effect may partly reflect age and career stage, since younger employees are more likely to be single.



5.8 Equity and career level help anchor employees

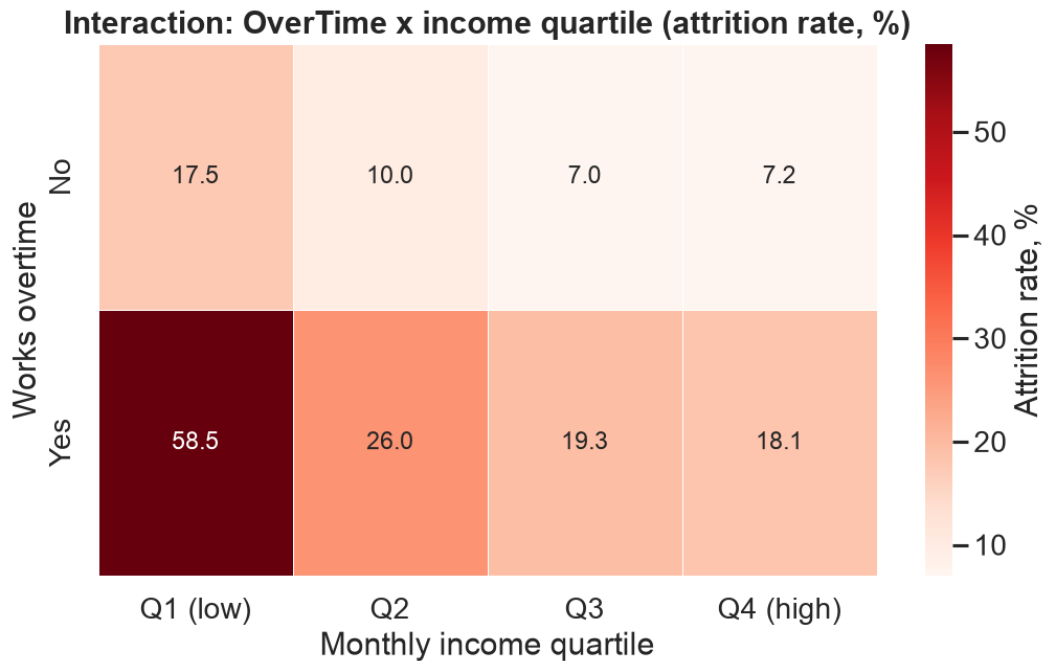
Stock options and job level both appear to protect retention. Employees with **no stock options** have a much higher attrition rate at **24.4%**, while employees with stock option levels **1–2** leave at only about **8–9%**. Job level shows the same pattern: entry-level employees have the highest attrition rate at **26.3%**, while levels **4–5** are below **8%**. This suggests that financial incentives and a clear career ladder both act as retention anchors, while employees without equity or at lower job levels are much more likely to leave.

5.9 Gender makes no difference (the rejected hypothesis)

Men leave 17.0% of the time, women 14.8%. A two-proportion z-test (the A/B test for two rates) gives $p = 0.26$, well above 0.05, so the gap is just sampling noise. This is a useful negative result: acting on gender would be both ineffective and a discrimination risk.

5.10 Risk factors compound

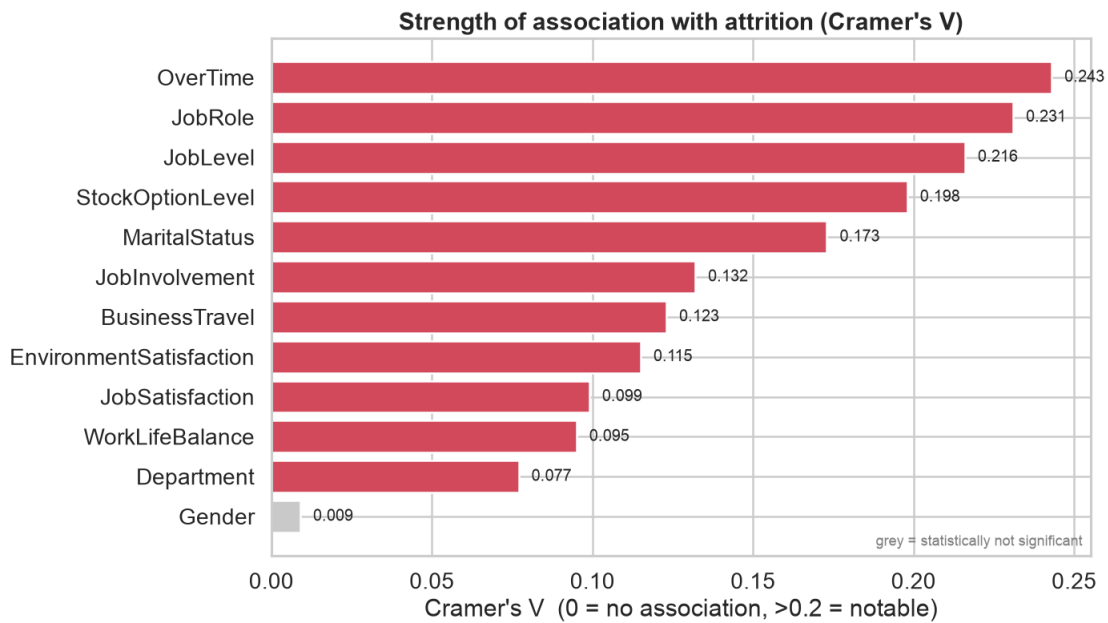
Combining two risks is worse than either alone. The worst segment - overtime plus the lowest income quartile - reaches 58.5%, versus 7.2% for the safest segment, roughly an eight-fold spread. HR should target combinations of risk, not single factors.



Overtime x income quartile: the factors multiply.

6. Which factors matter most

Ranking every categorical factor by Cramér's V (effect size) lines up with the findings above. Gender is the only factor that is not statistically significant.



The same numbers as a table:

Factor	Cramer's V	Significant (p<0.05)
OverTime	0.243	True
JobRole	0.231	True
JobLevel	0.216	True
StockOptionLevel	0.198	True
MaritalStatus	0.173	True
JobInvolvement	0.132	True
BusinessTravel	0.123	True
EnvironmentSatisfaction	0.115	True
JobSatisfaction	0.099	True
WorkLifeBalance	0.095	True
Department	0.077	True
Gender	0.009	False

7. Recommendations for HR

Priority	Action
High	Cap and compensate overtime; rebalance overloaded teams
High	Structured 0-12 month onboarding (mentor, 30/60/90-day check-ins, career path)
High	Benchmark and adjust pay for Sales Reps, Lab Technicians, entry-level roles
Medium	Extend stock options to critical junior / front-line roles
Medium	Pulse surveys on satisfaction and work-life balance; act on scores of 1-2
Targeting	Composite risk score (overtime + low pay + tenure < 1y + no options)

Where to start: the clearest quick win is the overtime + low-income group at 58.5%. Pulling that one segment toward the company average would save a meaningful amount of rehiring and retraining cost.

8. Limitations

- The data is synthetic and a single snapshot, so findings are associations, not proof of cause.
- PerformanceRating was unusable (only values 3 and 4).
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9. Technologies and tools used

Python

The whole analysis runs in Python. The libraries and what each was used for:

- pandas - loading the CSV, cleaning, feature engineering, and all the aggregations (groupby and pivot_table). Example: the attrition rate per group is just a grouped mean of a 0/1 flag.

```
df["AttritionFlag"] = (df["Attrition"] == "Yes").astype(int)
df.groupby("OverTime")["AttritionFlag"].mean() * 100 # attrition rate, %
```

- numpy - small numeric helpers, e.g. building TenureRatio safely (avoiding division by zero).

```
df["TenureRatio"] = np.where(df["TotalWorkingYears"] > 0,
                             df["YearsAtCompany"] / df["TotalWorkingYears"], 0)
```

- matplotlib + seaborn - all the charts: bar charts for rates by group (overtime, tenure, age, satisfaction, roles), box plots and density (KDE) plots for income and age distributions, and heatmaps for the pivot tables and correlation matrix.

```
sns.barplot(data=rates, x="OverTime", y="attrition_rate_%")
sns.heatmap(pivot_ot_income, annot=True, fmt=".1f", cmap="Reds")
```

- scipy.stats - the statistics: chi-square test and Cramér's V for categorical factors, Welch's t-test and Mann-Whitney for income, point-biserial correlation, and a two-proportion z-test for the gender A/B test.

```
chi2, p, dof, exp = stats.chi2_contingency(pd.crosstab(df["OverTime"],
df["Attrition"])))
```

SQL / BigQuery

SQL is used for the first pass over the data (section 4 of the notebook): counting rows, checking the overall attrition rate, and grouping by department. In the notebook this runs on an in-memory SQLite database so it works with no setup; the same queries run on BigQuery. Example (attrition rate by department):

```
SELECT Department,
       COUNT(*) AS headcount,
       ROUND(100.0 * AVG(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END), 2) AS
attrition_rate_pct
FROM employees
GROUP BY Department
ORDER BY attrition_rate_pct DESC;
```

Jupyter Notebook

The analysis lives in a Jupyter notebook so the commentary, code, tables and charts sit together and can be read top to bottom (also works in Google Colab).